

Recruitment Pipeline Analytics Dashboard

End-to-End Enterprise Case Study & System Architecture

Tools & Tech Stack:	Power BI Desktop, Star Schema Data Modeling, Advanced DAX
Core Functions Used:	CALCULATE, LOOKUPVALUE, DATEDIFF, AVERAGEX
Focus Area:	Data Architecture, Pipeline Efficiency, Time-Intelligence Integration

Executive Performance Metrics

Total Applications	Total Hired	Conversion Rate	Avg Days to Hire
900	51	5.7%	43 Days

1. Background & Business Problem

Managed and consolidated comprehensive recruitment pipeline data across multiple partner companies and various job openings moving through a strictly defined six-stage lifecycle: **Applied** → **Screening** → **Interview 1** → **Interview 2** → **Offer** → **Hired**.

- **Data Fragmentation:** Addressed critical leadership concerns regarding historical pipeline data living in disconnected spreadsheets, which previously made it impossible to evaluate pipeline health, track candidate drop-offs, or maintain a single source of truth.
- **Operational Blindspots:** Solved a severe lack of visibility into department-specific hiring velocity, unmeasured average time-to-hire benchmarks, and completely undefined conversion bottlenecks across job roles.

2. Data Engineering & Architecture

- **System Design:** Architected a robust five-table relational database structure (*applicants*, *jobs*, *companies*, *applications*, and *application_stages*) modeling an enterprise-grade Applicant Tracking System (ATS).
- **Data Modeling:** Connected source tables into a clean, hybrid star/snowflake schema, utilizing the **applications** table as the central fact table linked directly to normalized dimension tables.
- **Relationship Management & Loop Prevention:** Optimized data flow by routing company relations strictly through jobs (*companies* → *jobs* → *applications*), preventing circular paths and eliminating ambiguous active model relationships.
- **Time-Intelligence Integration:** Developed and integrated a dedicated custom *DateTable* related to the applied date field to unlock seamless month-over-month trend analysis and time-based filtering.

3. Advanced DAX & Metrics Implementation

Engineered core volume and efficiency measures to standardize pipeline performance evaluation:

- **Volume Metrics:** Created *Total Applications* (overall volume received) and *Total Hired* (count of applications with a final stage status of 'Hired').
- **Conversion Rate %:** Developed a complex measure dividing *Total Hired* by *Total Applications* to establish a baseline efficiency percentage.
- **Avg Days to Hire:** Implemented time-intelligence and iterator calculations utilizing DATEDIFF and AVERAGEX to accurately compute precise time gaps from the initial application date to the final hiring stage.
- **Conversion Rate by Department:** Configured row-level evaluation contexts to slice conversion efficiencies dynamically across engineering, operations, customer support, data, and finance teams.

4. Dashboard Components & Visual Architecture

- **Executive KPI Row:** High-level summary cards displaying core metrics (900 Applications, 51 Hires, 5.7% Conversion, 43 Days Avg Time-to-Hire) to give immediate operational context.
- **Recruitment Funnel Visual:** Custom funnel chart mapping precise candidate counts and drop-offs across all six sequential pipeline stages to visually isolate structural losses.
- **Trend & Breakdown Visuals:** Built a Monthly Applications Trend line chart displaying volume and seasonality; a Donut chart for Final Status breakdown (*Rejected*, *Withdrawn*, *Hired*, *In Progress*); and horizontal Bar charts isolating applications by job title and company.
- **Interactive Filtering:** Integrated interactive cross-filtering slicers for Department, Company Name, and Date Range to empower stakeholders with custom, self-service data exploration.

5. Key Business Insights & Recommendations

- **Funnel Bottleneck:** Discovered a major operational roadblock between **Interview 1** and **Interview 2**, where candidate count crashes sharply from 487 down to 254 (representing a massive **~48% drop-off loss** at this single transition).
- **Outcome Concentration:** Highlighted that **81.8%** of total applications ultimately end in a 'Rejected' status, while maintaining an average time-to-hire benchmark of **43 days**.
- **Strategic Interventions:** Recommended immediate qualitative reviews of interviewer feedback and screening criteria at the Interview 1 to Interview 2 transition phase.
- **Efficiency Targets:** Proposed setting strict time-to-hire target thresholds (e.g., maximum 30 days) and systematically sharing high-converting departmental best practices across underperforming teams.